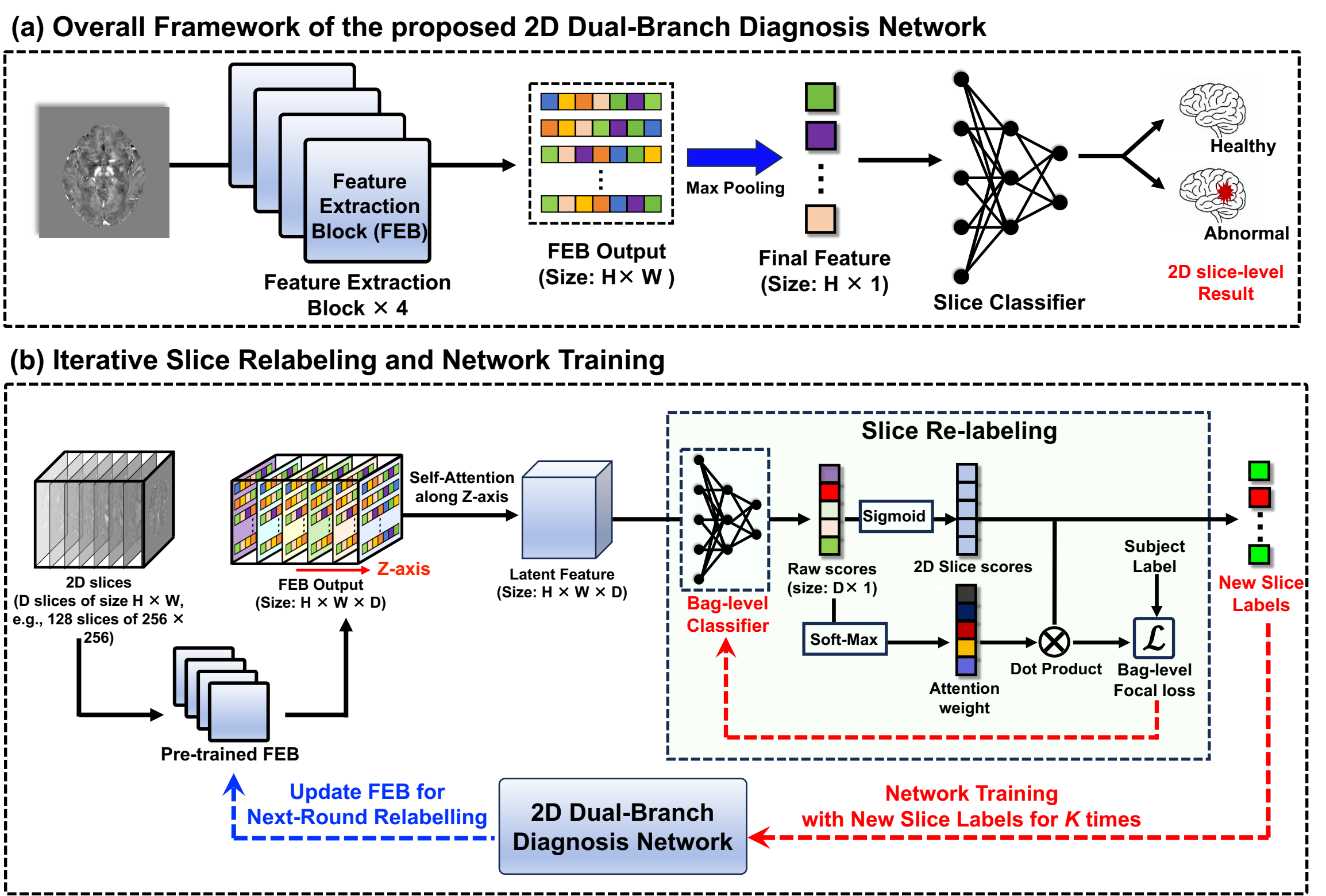

(a) Overall Framework of the proposed 2D Dual-Branch Diagnosis Network
Feature Extraction Block (FEB)
Feature Extraction Block × 4
FEB Output (Size: H× W )
Max Pooling
Final Feature (Size: H × 1)
Slice Classifier
Healthy
Abnormal
2D slice-level Result
(b) Iterative Slice Relabeling and Network Training
2D slices (D slices of size H × W, e.g., 128 slices of 256 × 256)
Pre-trained FEB
FEB Output (Size: H × W × D)
Z-axis
Self-Attention along Z-axis
Latent Feature (Size: H × W × D)
Slice Re-labeling
Bag-level Classifier
Raw scores (size: D× 1)
Sigmoid
2D Slice scores
Soft-Max
Attention weight
Dot Product
Subject Label
Bag-level Focal loss
New Slice Labels
Update FEB for Next-Round Relabelling
2D Dual-Branch Diagnosis Network
Network Training with New Slice Labels for K times